

(d) Collecting data

Students should be taught to:

collect or obtain data by observation, surveys, experiments (including controlled experiments), counting, data logging, questionnaires and measurement;

obtain primary data by questionnaires or experiment;

understand the effects of accuracy on measurements;

Writing improved or good questions for a questionnaire will be expected.

Knowing that measured data such as length or time is subject to some error. For example, that every measurement is taken to a given level of accuracy.

Content

understand the advantages and disadvantages of using interviews versus questionnaires;

design and use efficient and effective data capture sheets and methods of recording data;

understand the role, and use of, pilot studies and pre-testing;

understand and account for the problems of design, ambiguity of wording, leading and biased questions, definitions and obtaining truthful responses;

understand the advantages and disadvantages of open and closed questions;

be aware of and understand the problems related to identifying the appropriate population, the distribution and collection of questionnaires, errors in recorded answers, non-responses and missing data;

identify appropriate sources of secondary data;

extract data from secondary sources, including those based on ICT;

understand the aspects of accuracy, reliability, relevance and bias as related to secondary data;

design simple statistical experiments to obtain data;

understand the meaning of explanatory and response variables;

understand the need for the identification of the variables to be investigated;

understand surveys.

Notes

Deciding which technique might be more appropriate, and why, will be expected.

The rationale behind pilots and pre-tests will be expected.

The minimisation of ambiguity and bias will be expected.

Incorporated in questionnaire design.

Dealing with problems such as non-response and rogue values will be expected.

Newspapers, national statistics, the internet and others.

The sampling of secondary data from sources such as National Statistics will be expected or data on subjects of students' own interests, including that extracted from the internet.

Questioning the reliability of secondary sources and data will be expected.

Examples of secondary data include the internet, RPI, Key Data and Abstract of Statistics, GCSE results, etc.

Students will be expected to comment on the design of experiments, eg using controls and random allocation.

The identification of explanatory (independent) and response (dependent) variables will be expected.

Knowledge of redundant variables will be expected.

Examples from other school subjects (including science) and everyday life.